Drift-Sense

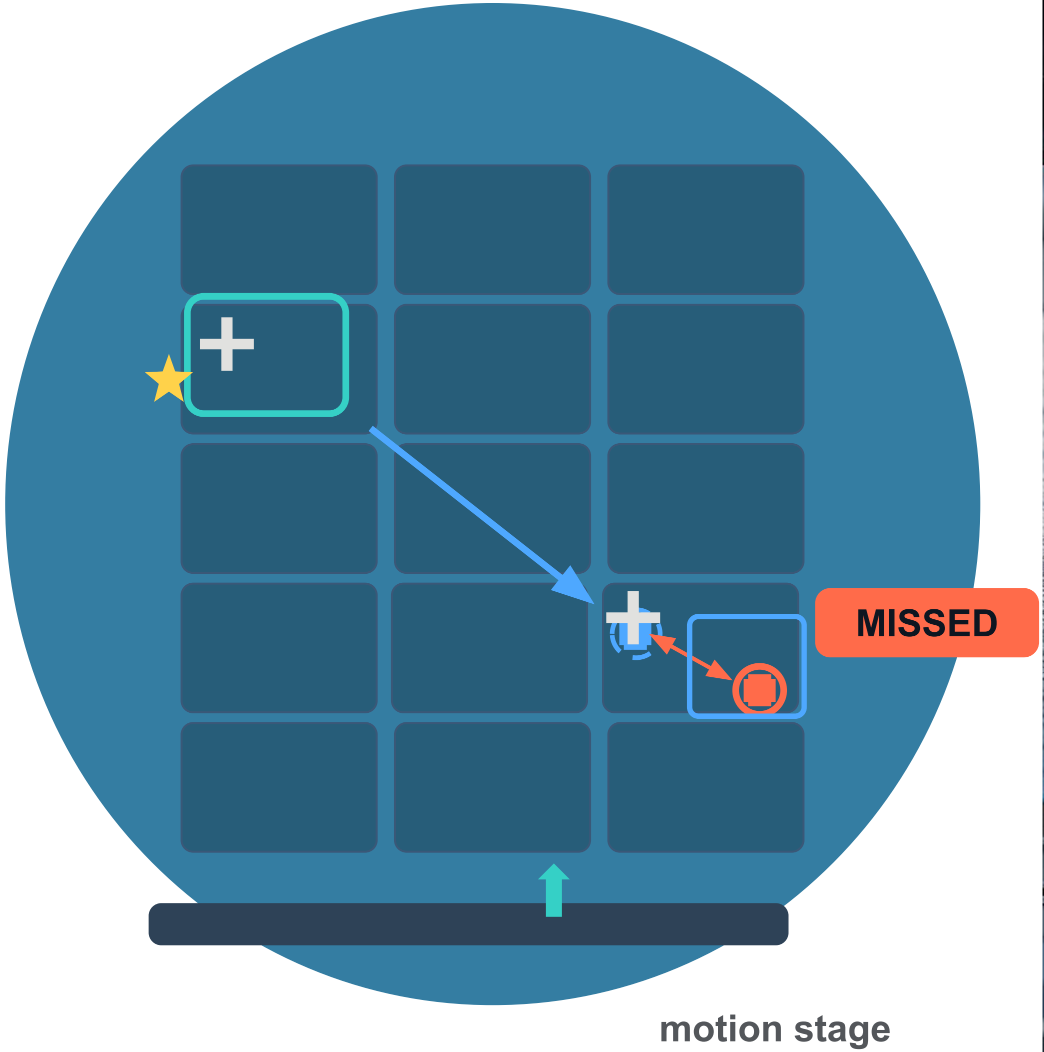
Finding a Needle in Nanoscale Haystack

(Navigation-Error Recovery)

Background

1. Dies repeat across the wafer — they look nearly identical.
2. A site was characterized here earlier — the reference location.
3. The tool must grab at the 'same' site on another die.
4. The motion stage accumulates tiny errors — drift, vibration, thermal effects.

- ★ Reference site (star)
- Ideal target (dashed crosshair)
- Actual landing (solid crosshair)



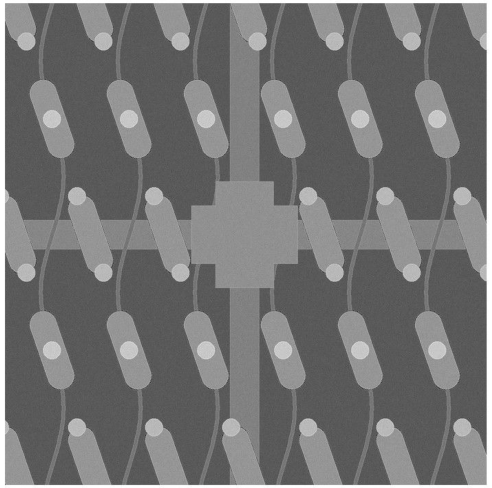
Description

Let’s say we have 2 Optics zoom 10x and 100x

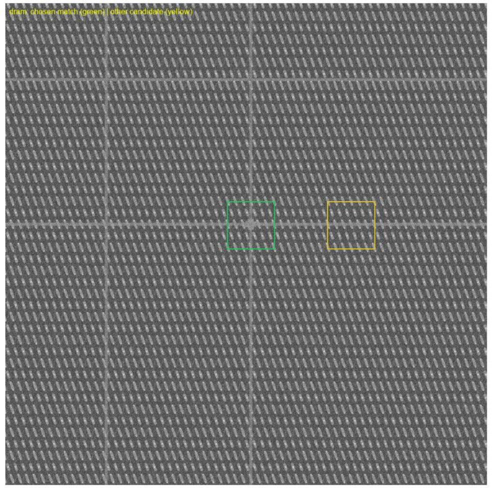
Reference Image is always zoomed-in (100x) image of Target location

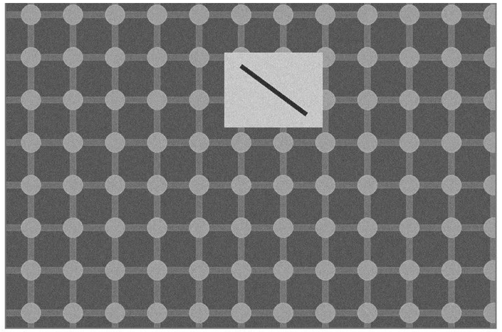
Search Image will be a zoomed-out image(10x) of the Target location

Reference (1000x1000, 1nm/px, DRAM staggered array)

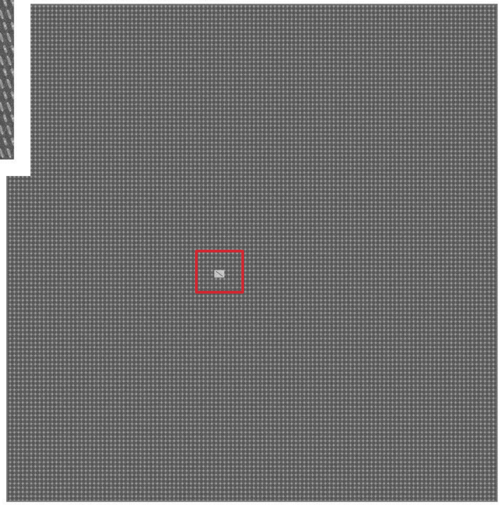


Wide search (10nm/px, noisier) w/ bbox





e search (1000x1000 px, 10 nm/px)



APPLIED MATERIALS

Description

1. Find zoomed-in reference image in search Image.

2. Provide center x, y of the first matching tile

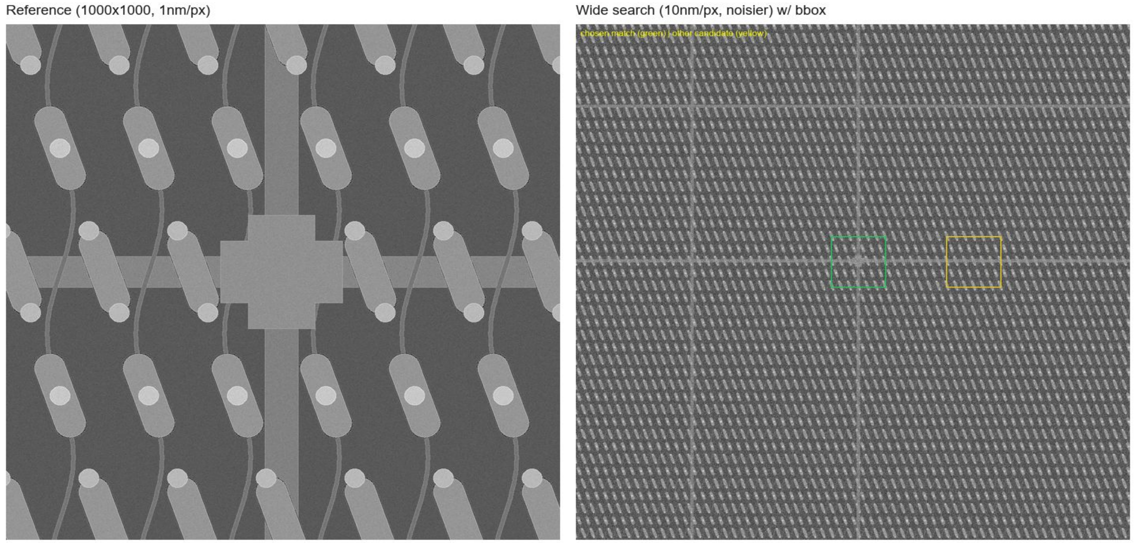
3. In case more than one tile find closest to the reference image center.

**This is a search-and-localize problem: finding a known pattern, at a known but reduced scale, inside a noisier, lower-resolution image that may also contain visually similar repeating structures — since DRAM- and FinFET-style layouts are themselves highly periodic, which is what makes this genuinely hard rather than a simple exact-pixel lookup.*

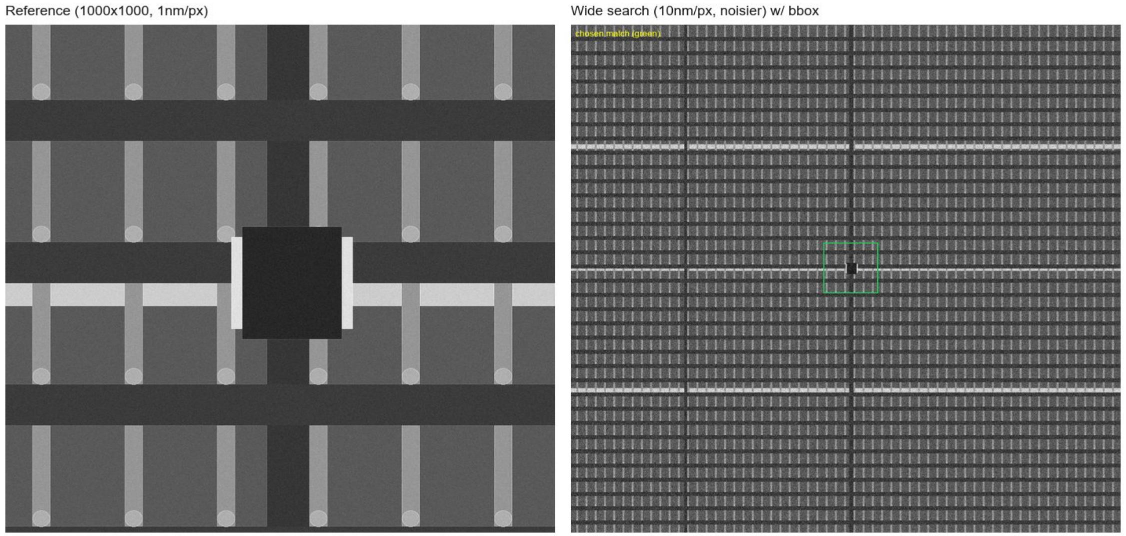
Parameters and Datasets

- Both the reference and wide-search images are grayscale, 1000×1000 pixels.
- At the high-resolution ("100x") capture, pixel size is 1 nm — a 1 µm × 1 µm field of view.
- The wide-search ("10x") capture covers exactly 10x the physical area at the same pixel count (10 nm/pixel, ~10 µm × 10 µm field of view) — the reference pattern appears shrunk by that same 10x factor somewhere inside it.

* Due IP constraints, No dataset is provided; participants shall generate their own synthetic image-pairs.



Example 1: A sample generated Reference (left) and the same pattern shrunk 10x somewhere inside the wide search image (right).



Example 2: A sample generated Reference (left) and the same pattern shrunk 10x somewhere inside the wide search image (right).

APPLIED MATERIALS

Expected Solution:

The solution should:

- **Build a synthetic-but-realistic grayscale** dataset generator producing reference and wide-search image pairs mimicking either DRAM-style (periodic memory-cell / word-line / bit-line / contact arrays) or FinFET-style (parallel fin / gate structures) die architecture — participant's choice, judged equally either way — using only publicly known structural characteristics, never proprietary fab data.
- Justify **every augmentation/noise choice**, distortion, rotation & scaling against at least 2–3 credible public sources (papers, textbooks, or patents on semiconductor device structure or SEM imaging), cited in the final presentation.
- Correctly account for the true 10x zoom ratio between the high-resolution reference and the wide-search image, **then use a classical ML or DL-based** localization algorithm of your choice to find the reference pattern and **report the center (x, y)** of the matching region — or, if more than one region matches, whichever is closest to the search image's center.
- Report a measurable success rate: **1) computation time of algorithm** on single image (1k*1k), 2) run the method across at least **30 randomized generated test cases** and report the percentage landing within a stated tolerance (e.g., within subpixel of the true down sampled location), plus at least one honest example of where it fails (e.g., inside a highly periodic array region) and why.
- **Bonus**: Solutions that additionally generalize to **optical microscope images (RGB, 3-channel**) not just the primary SEM grayscale case — earn bonus credit, provided the core SEM-based solution is completed first.

FAQs

- What is the image size
  - 1000*1000 pixels for both search and reference image.

- What is the submission format ?
  - Zip file containing
    - Pip freeze your env dependencies to reproduce the exact results.
    - Well documented Python file generating the dataset you created including noise modelling methods.
    - Well documented Python file generating the center x,y results on a given pair of images 1k*1k (search, reference)
    - DL model and training python notebook (if DL methods are used).
    - Any supporting documents for methods or citations

- Can we submit code in c++ or one single jupyter notebook ?
  - No. It will be difficult to evaluate different languages. Please use generative AI to migrate code to python

- Which noise models should be added to these images ?
  - Please refer to online literature for SEM Image noise models. You should assume wide-search image will be more noiser in test data to check algorithm robustness.

FAQs

- Do we have sample data for training ?
  - We are sharing starter prompt to generate sample data.
  - Participants should look for web articles and create real-like search datasets.
  - We will share later a github repo for some code samples to generate more test data as per event schedule.

- How is final scoring done ?
  - 50% of marks on inference results on finding correct coordinates of the pattern on test data (includes computation time)
  - 30% of marks on Augmentation code which can create real-like SEM images of FinFET/DRAM stack based on literature study,
  - 10% on identifying the root cause or explainability on failure cases.
  - Bonus marks if same can be done RGB images of Optical Tools

- Detailed Evaluation Rubic will be shared?
  - Yes, we will share it later with the test dataset.

Sample Prompt to Generate data

1. *Write Python code (PIL/numpy) that generates one example pair of images for a wafer-inspection navigation-error problem:*
2. *A "reference" image — grayscale, 1000×1000 pixels, representing a small high-resolution capture of a semiconductor die region. Choose one style:*
   - *DRAM-style: a periodic array of horizontal word lines and vertical bit lines crossing at right angles, with a small contact/via dot at every intersection.*
   - *FinFET-style: a dense set of parallel vertical fin lines, crossed by one or two horizontal gate bars.*
3. *A "wide search" image — same 1000×1000 pixel size, but representing exactly 10x the physical field of view of the reference. Build this by generating a larger continuous version of the same die layout (so the pattern repeats naturally across it, the way real dies do), then resizing that larger region down t o 1000×1000 to represent the coarser resolution. The reference pattern should appear somewhere inside this wide image, shrunk down by that same 10x factor.*
4. *Add realistic touches: independent grayscale sensor noise on each image (don't reuse the same noise on both — they're t wo separate captures), and a bit of edge-brightening to mimic how real SEM images show brighter contrast along feature e dges.*
5. *Save both images and print their sizes, so I can visually confirm the reference pattern is genuinely visible (shrunk) inside th e wide image before I build anything more sophisticated on top of it.*

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